

Lecture 17

Object detection with R-CNN, SSD, and YOLO

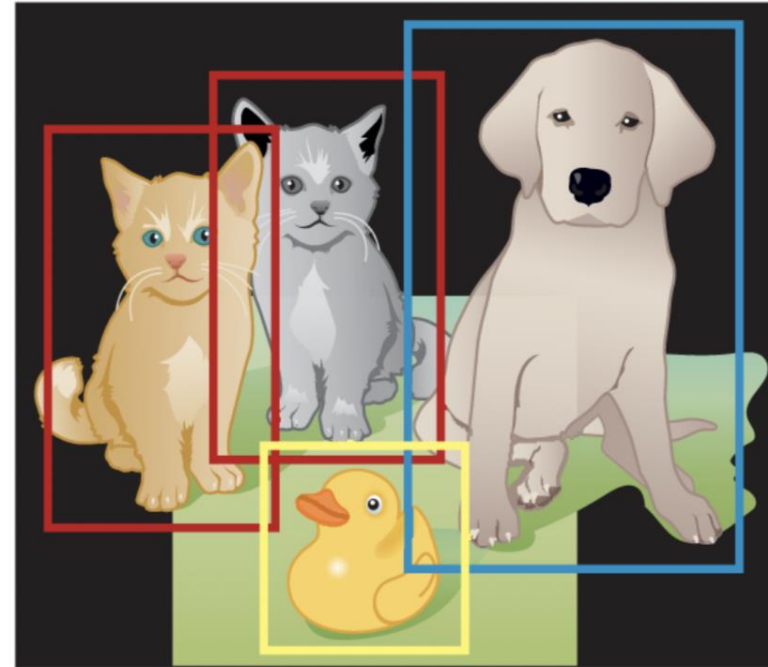
Classification Vs Object Detection

Image classification



Cat

Object detection
(classification and localization)



Cat, Cat, Duck, Dog

Classification Vs Object Detection

Image classification	Object detection
The goal is to predict the type or class of an object in an image. ■ Input: an image with a single object ■ Output: a class label (cat, dog, etc.) ■ Example output: class probability (for example, 84% cat)	The goal is to predict the location of objects in an image via bounding boxes and the classes of the located objects. ■ Input: an image with one or more objects ■ Output: one or more bounding boxes (defined by coordinates) and a class label for each bounding box ■ Example output for an image with two objects: – box1 coordinates (x, y, w, h) and class probability – box2 coordinates and class probability Note that the image coordinates (x, y, w, h) are as follows: $(x$ and $y)$ are the coordinates of the bounding-box center point, and $(w$ and $h)$ are the width and height of the box.

General object detection framework Components

Region Proposal

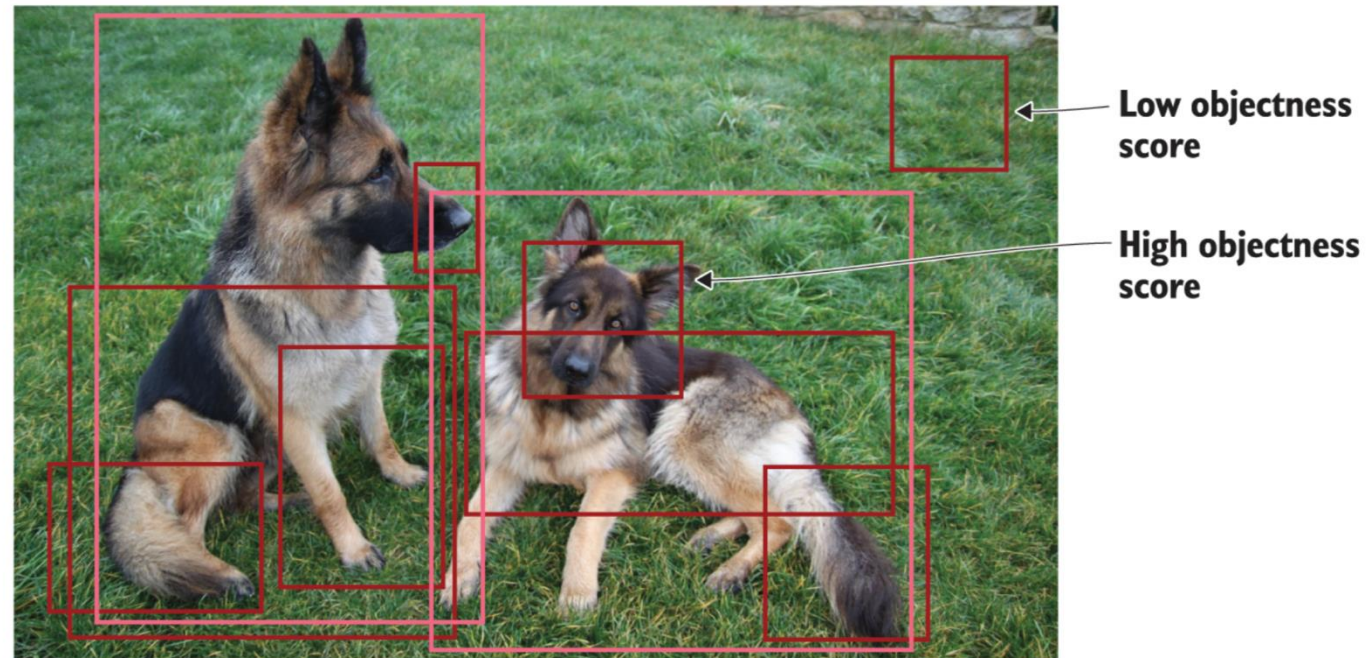
Feature extraction and
network predictions

Non-maximum
suppression (NMS)

Evaluation metrics

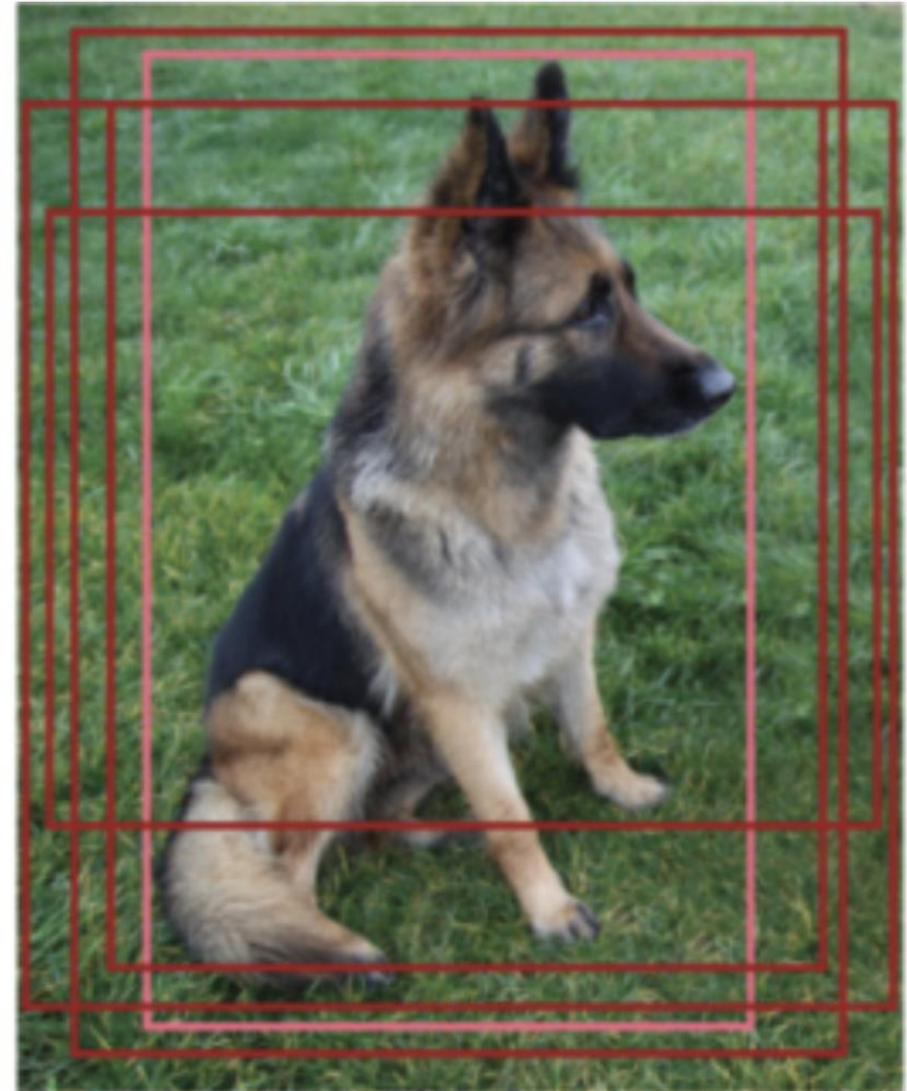
Region Proposals

- Assigns objectness score
- Region proposal generation
 - Selective search algorithm
 - Complex visual feature extraction using deep neural networks
- Produces a number of bounding boxes for analysis
 - regions are accepted/rejected based on the objectness score
- Computationally expensive



Network Predictions

- Uses image classification model to extract visual features
- Predictions
 - Bounding box
 - A tuple $\{x,y,w,h\}$
 - $\{x,y\}$ – bounding box center point coordinates
 - w – box width
 - h – box height
 - Class corresponding to the object in the image
- Can predict multiple bounding boxes for the same object if multiple RoIs are predicted from the region proposal step



Non-maximum suppression (NMS)

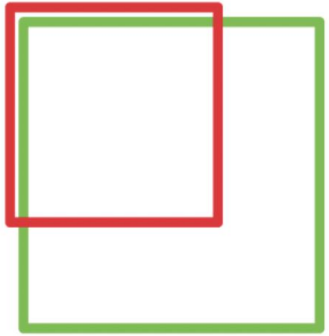
- Finds a single bounding box for the object in the image
- Checks the object's maximum prediction probability from the surrounding boxes
 - Eliminate the bounding boxes comparing the predictions with the confidence threshold
 - Select the bounding box with the highest probability from remaining boxes
 - Check for the overlap between remaining boxes with same class predictions
 - Use Intersection over Union (IoU) to average out the bounding boxes with high overlap
 - Eliminate the boxes with IoU value less than NMS threshold
- Hyperparameters to tune: confidence threshold and NMS threshold

Object Detection Evaluation Metrics

- FPS: frame per second
 - Measures object detection speed
- mAP: Mean Average Precision
 - Better object detection for higher values
- Intersection over Union (IoU)
 - Measure the overlap between 2 bounding boxes

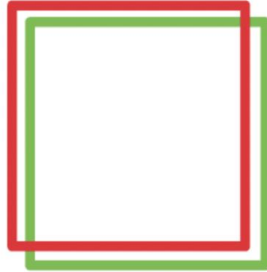
Intersection Over Union (IoU)

IoU: 0.4034



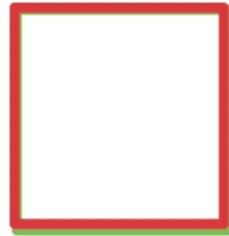
Poor

IoU: 0.7330



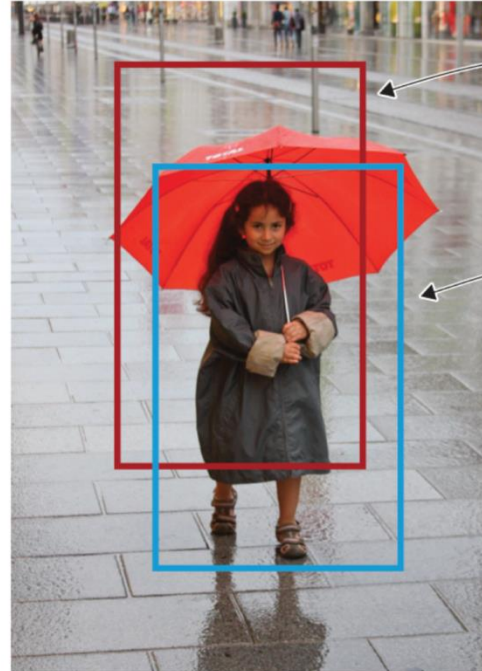
Good

IoU: 0.9264



Excellent

- $$IoU = \frac{B_{groundtruth} \cap B_{predicted}}{B_{groundtruth} \cup B_{predicted}}$$



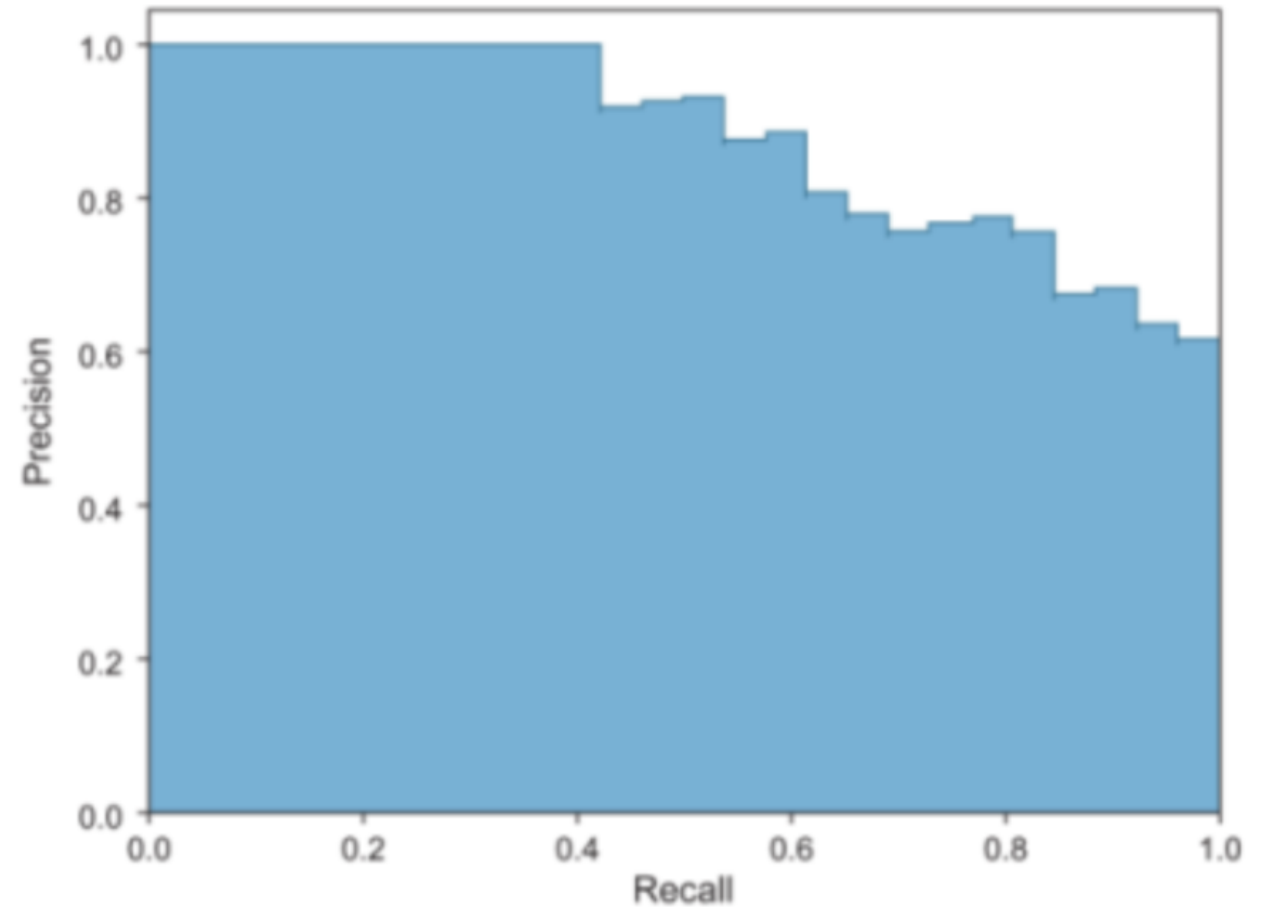
Predicted person bounding box

Ground truth person bounding box

$$\text{Score} = \frac{\text{Area of overlap}}{\text{Area of union}}$$


Precision-recall curve (PR curve)

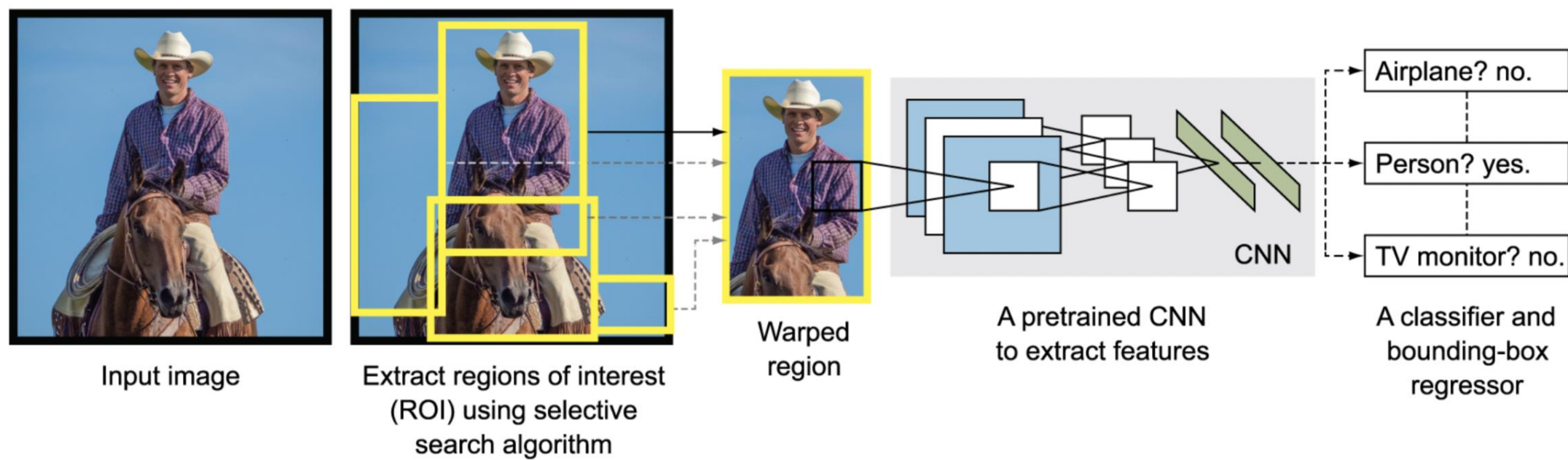
- $Recall = \frac{TP}{TP+FN} = \text{Sensitivity}$
- $Precision = \frac{TP}{TP+FP} = \text{Specificity}$
- Precision and Recall values for a detector should stay high irrespective of change in the confidence threshold.



mAP (Mean Average Precision)

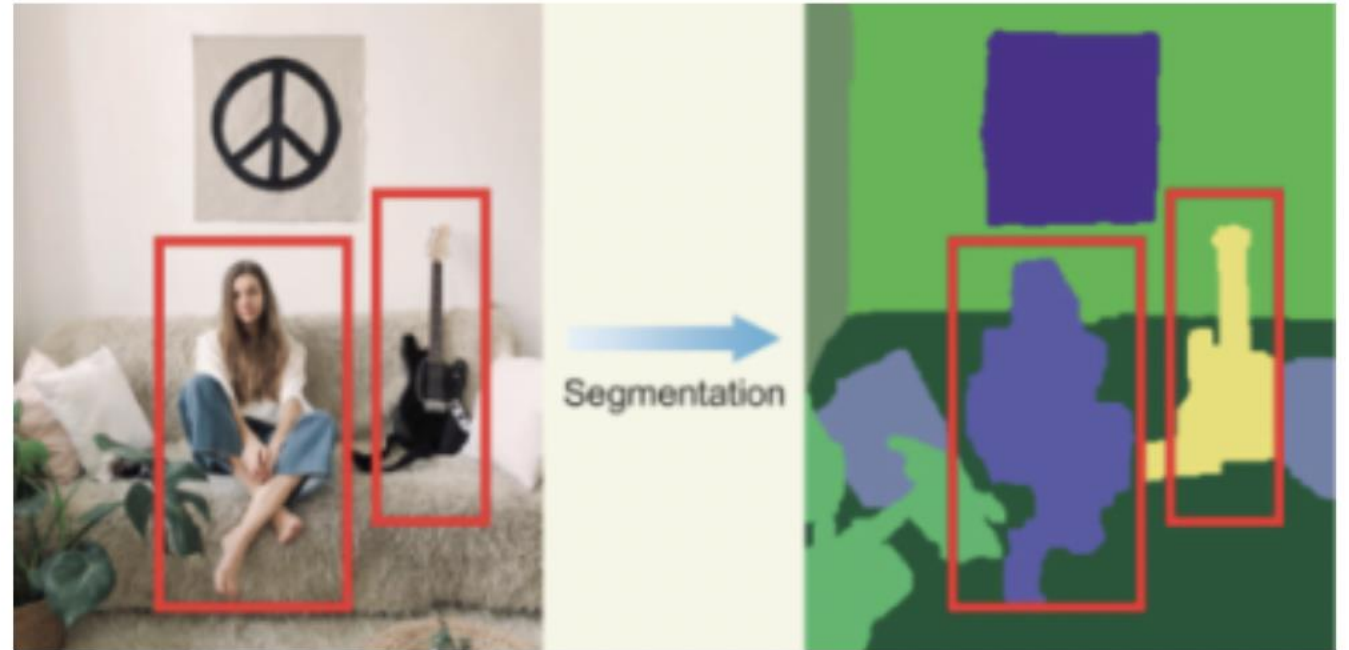
- Each bounding box will have an objectness score associated (probability of the box containing an object).
- Precision and recall are calculated.
- Precision-recall curve (PR curve) is computed for each class by varying the score threshold.
- Calculate the average precision (AP): it is the area under the PR curve. In this step, the AP is computed for each class.
- Calculate mAP: the average AP over all the different classes.

R-CNN

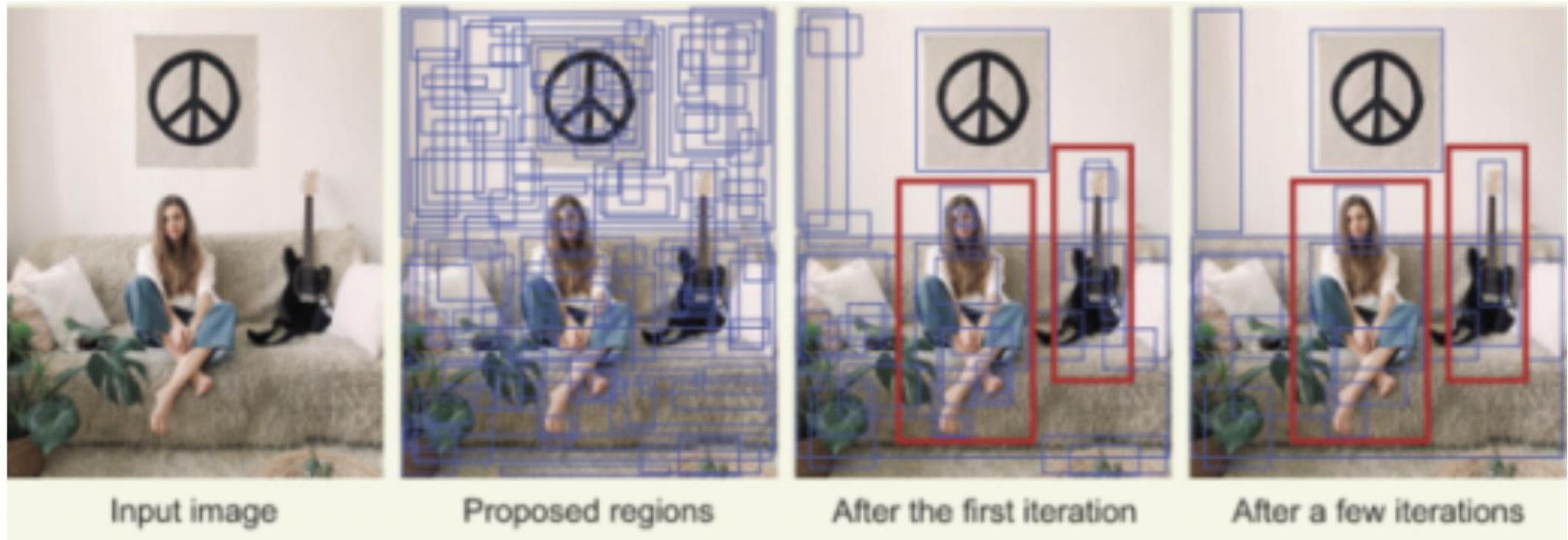


Selective Search Algorithm

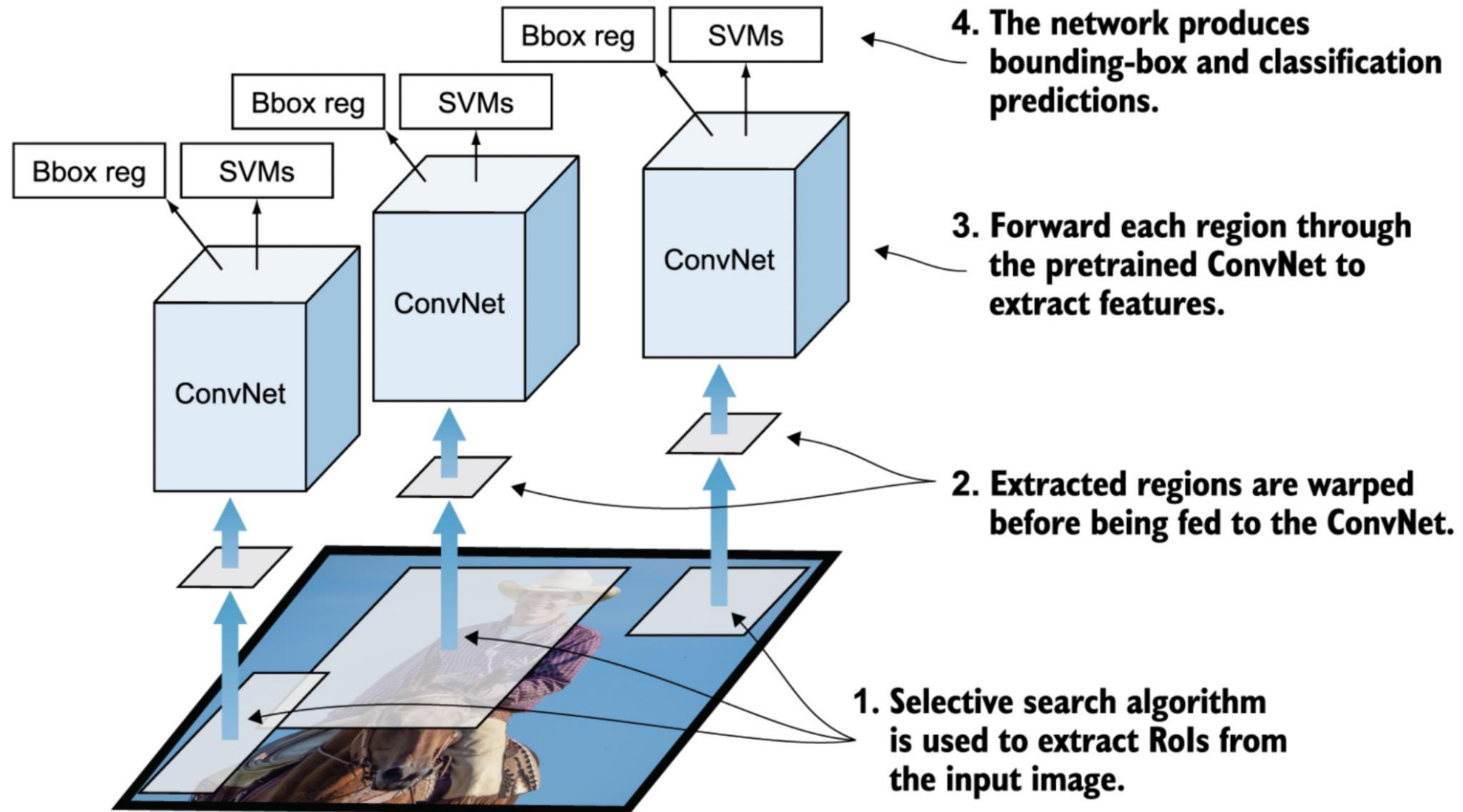
- Greedy search algorithm
- Finds an object in the images by combining similar pixels and textures into rectangular boxes
- Applies segmentation algorithm to find blobs in the image



Bottom-up segmentation using the selective search algorithm



R-CNN Architecture



Training R-CNNs

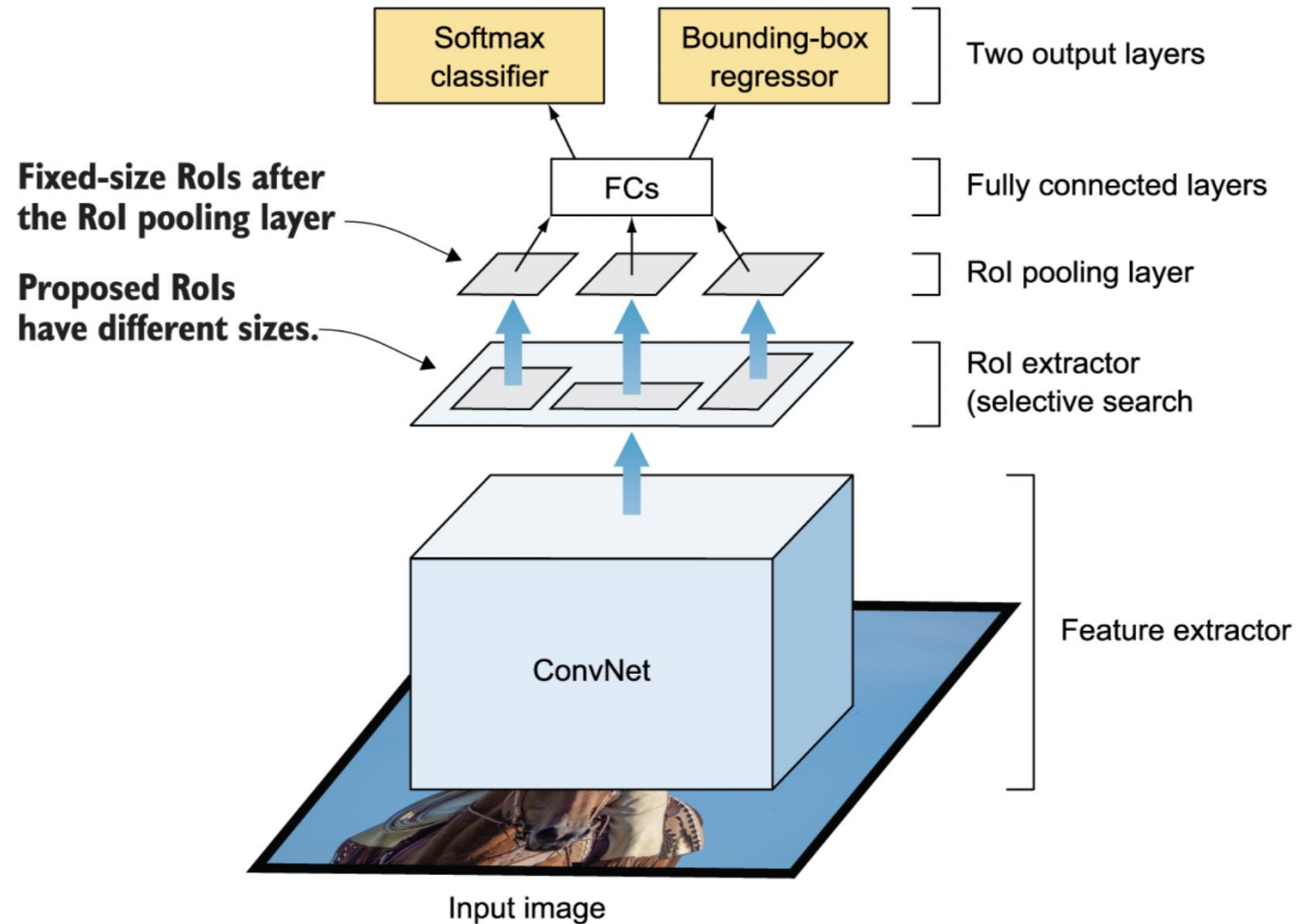
- Modules in R-CNNs
 - Region proposal
 - Feature extractor
 - Classifier
 - Bounding-box regressor
- Trainable modules
 - Feature extractor - CNN
 - Classifier – SVM
 - Bounding-box regressor
- Slow training of three separate modules without any overlap

Disadvantages of R-CNNs

- Not a single end to end system
- Performs feature extraction for each of the proposed region – not suitable for real time applications
- Multi-stage training pipeline
 - Expensive in terms of space and time

Fast R-CNN

- Improved detection speed and accuracy
- Applies CNN feature extracted on the input image before the region proposal algorithm
- Uses softmax layer instead of SVM for classification
- Uses selective search algorithm for region proposal
- RoI pooling layer – extracts a fixed-size window from the feature map



Multi-task loss function

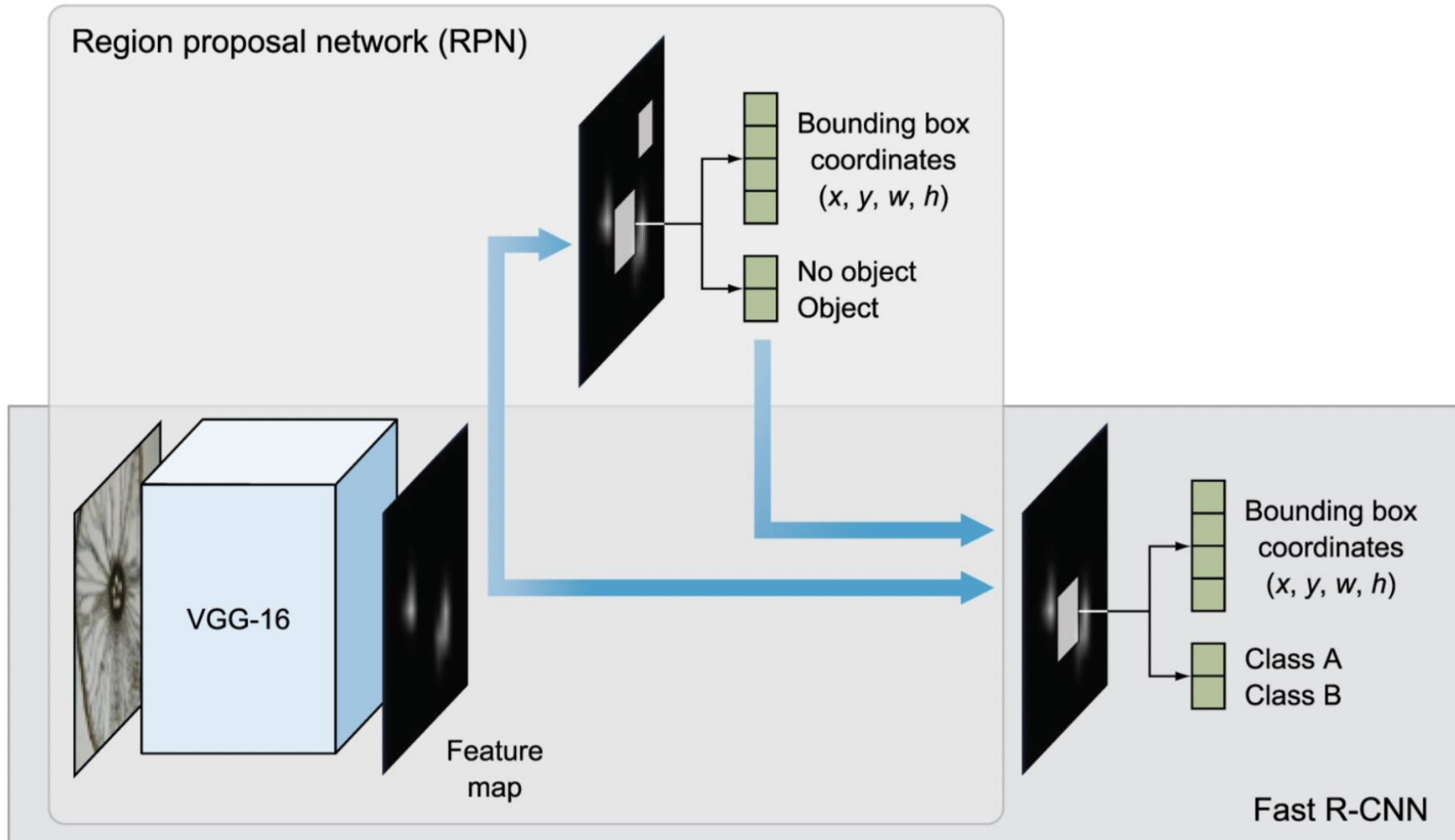
- Overall Loss function = Classification loss + Regression loss
- Classification loss, $L_{cls}(p, u) = -\log p_u$ -> cross-entropy loss
- Regression loss, $L_{loc}(t^u, u) = \sum L1_{smooth}(t_i^u - v_i)$ -> absolute error

$$L(p, u, t^u, v) = L_{cls}(p, u) + [u \geq 1] l_{box}(t^u, v)$$

Disadvantages of Fast R-CNN

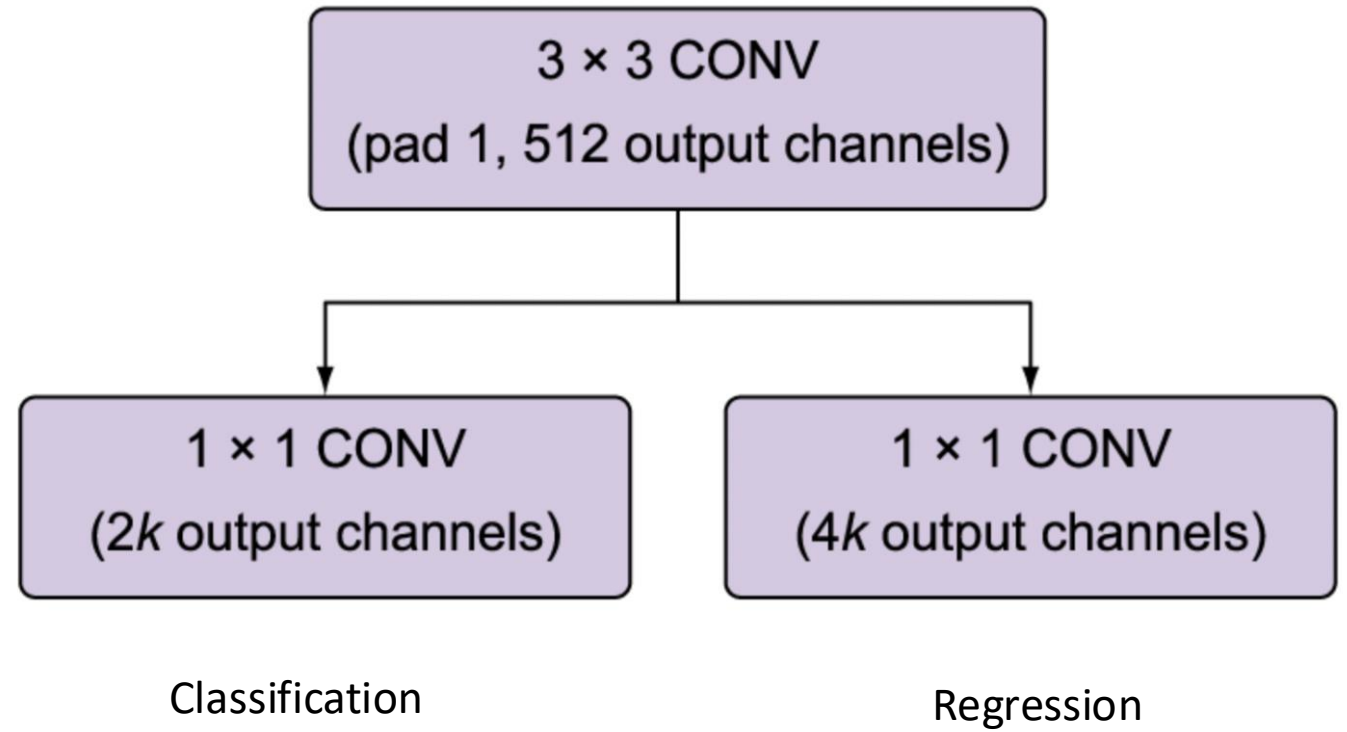
- Slower region proposal algorithm (selective search)
- Regions are proposed by a separate algorithm which is not a part of the object detection CNN

Faster R-CNN



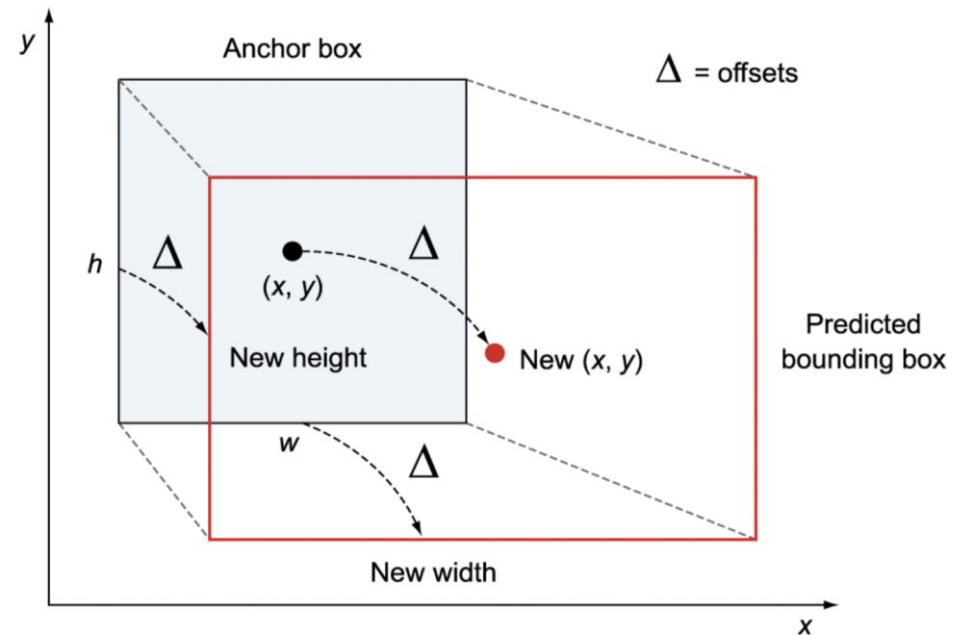
Region Proposal Network (RPN)

- Also known as attention network
- RPN classifier predicts the objectness score
- RPN regressor uses anchor boxes to predict the bounding box co-ordinates of the proposed regions

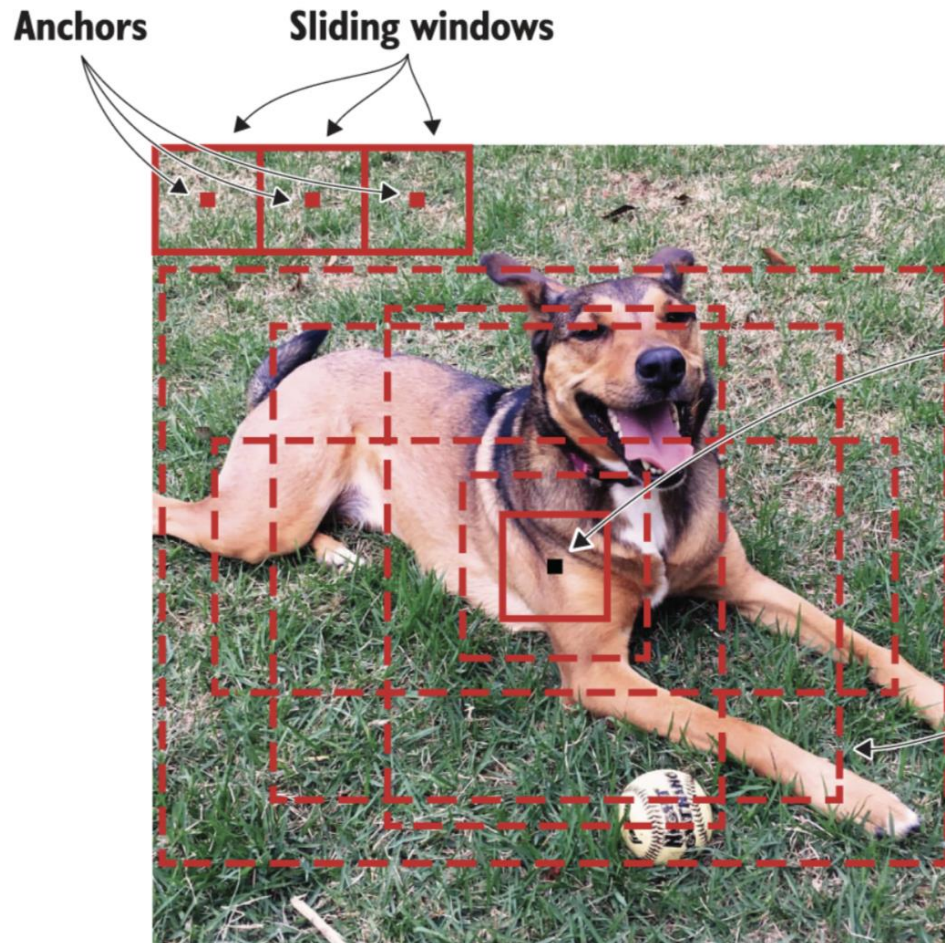


Anchor Boxes

- Challenging to enforce rules to predict the center points inside the image
- Solution: create anchor boxes and predict the offsets from these boxes



Anchor Boxes

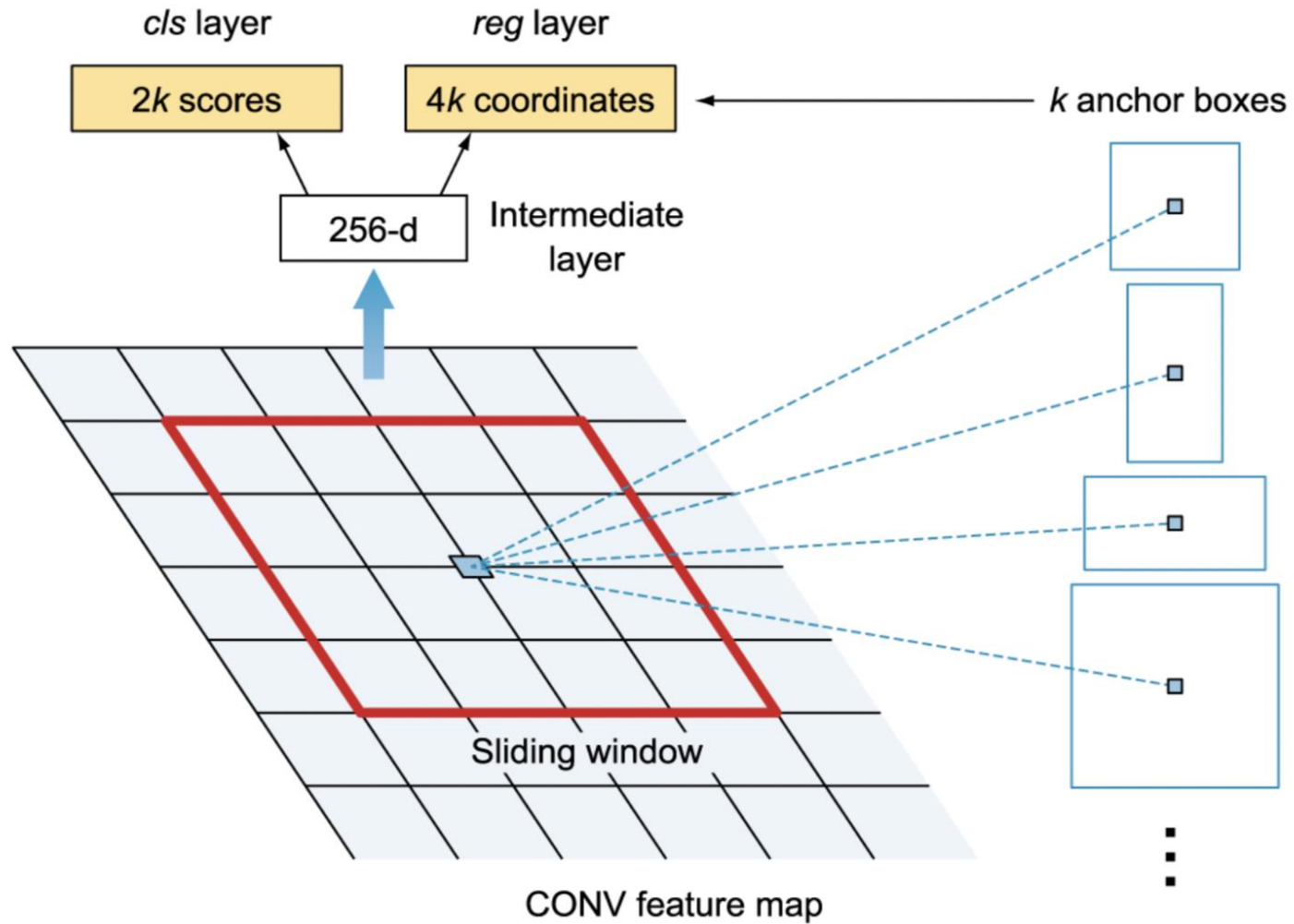


The anchor is placed at the center of the sliding window. Each anchor has anchor boxes with varying sizes.

The IoU is calculated to choose the bounding box that overlaps the most with the ground-truth bounding box.

Anchor boxes

Region Proposal Network (RPN)



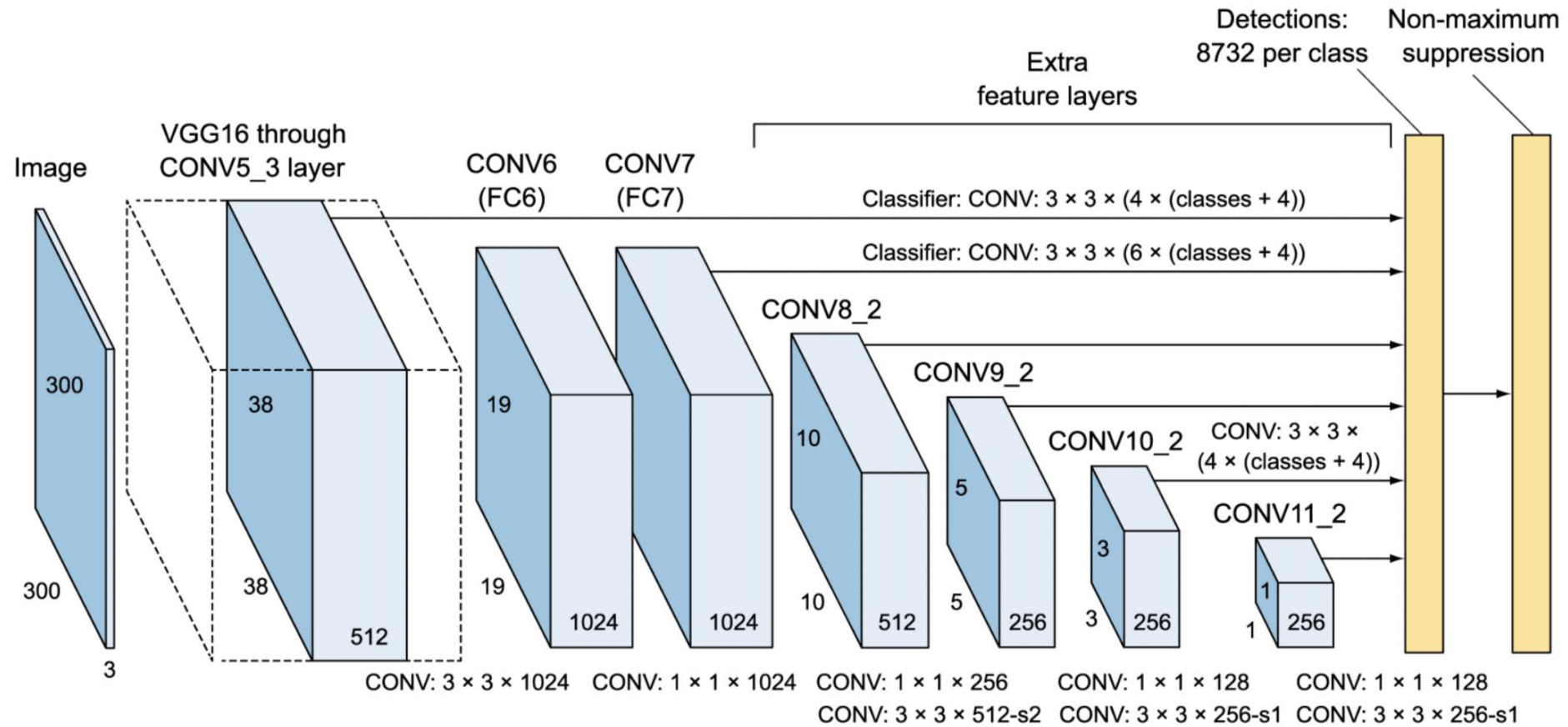
Limitations of Faster R-CNN

- Longer training time
- Training in multiple phases
- Slow at inference time

Single Shot Detector (SSD)

- Performance: 74% mAP at 59 FPS
- Skips the region proposal phase, runs detection directly over a dense sampling of possible locations
- Main components of SSD
 - Base network for Feature extractor
 - Multi-scale feature layers
 - Non-maximum suppression

SSD Architecture

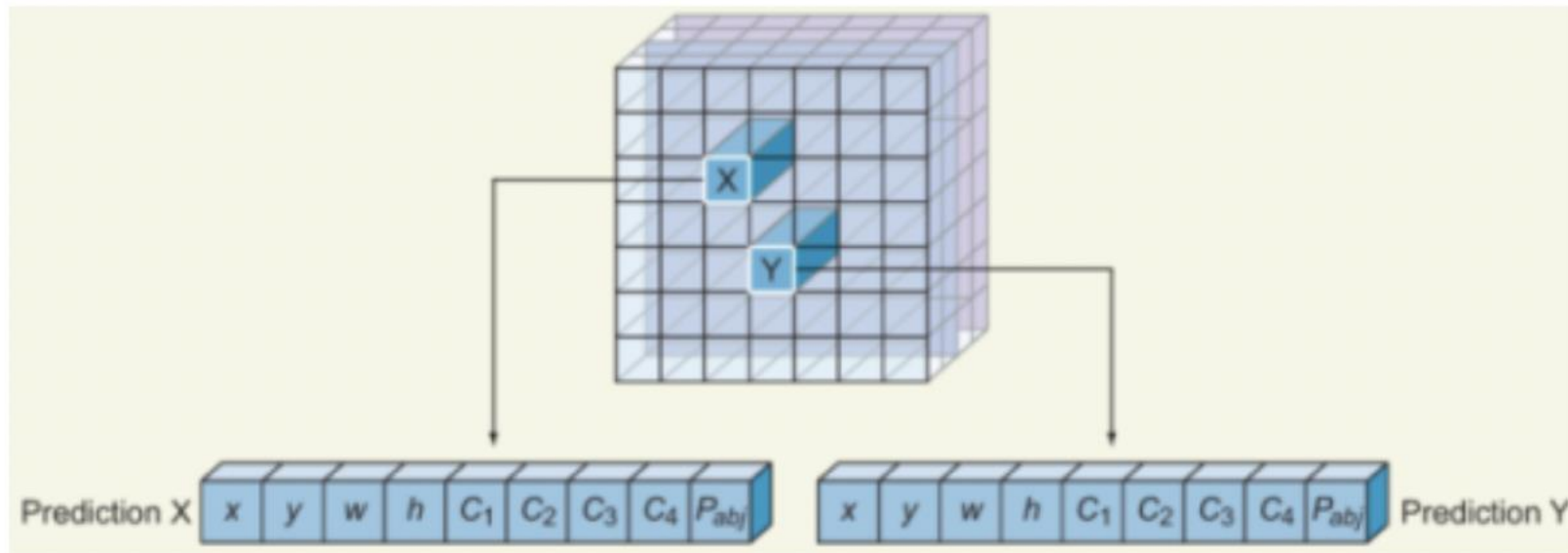


Output Prediction

- 4 values that describe the bounding-box (x, y, w, h)
 - + 1 value for the objectness score
 - + C values that represent the probability of each class.
- That's a total number of prediction values = $5 + C$.

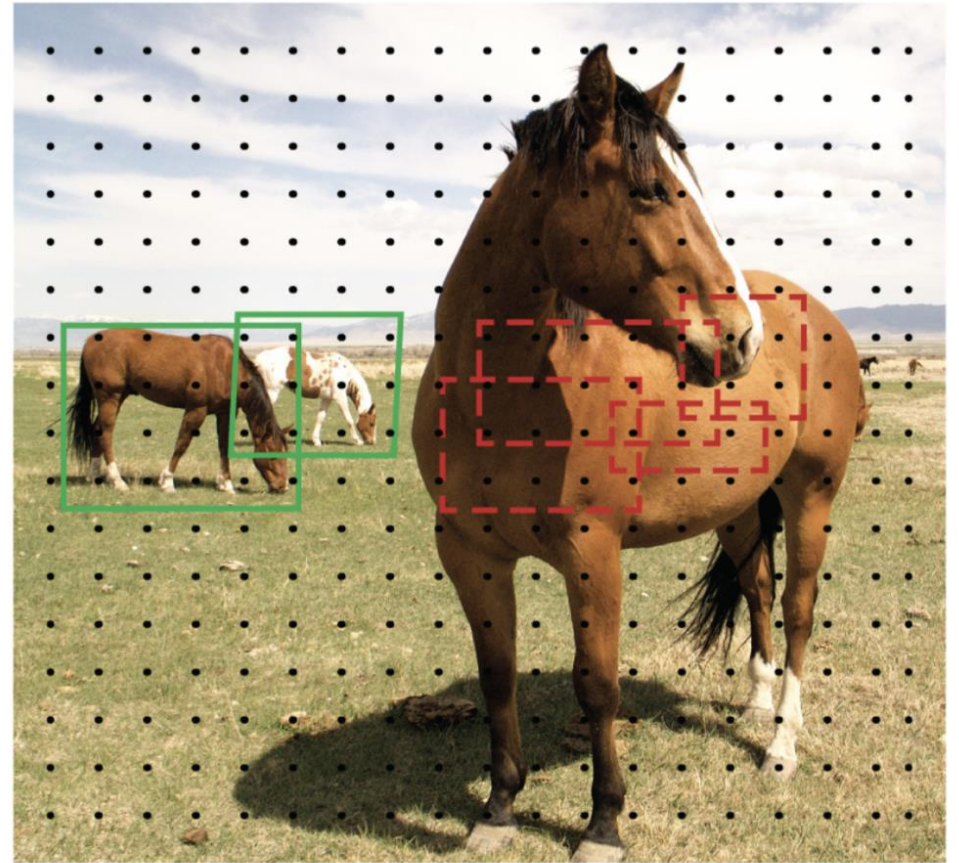
Suppose there are 4 object classes.

Then each prediction will be a vector that looks like this: $[x, y, w, h, \text{Objectness Score}, C_1, C_2, C_3, C_4]$.



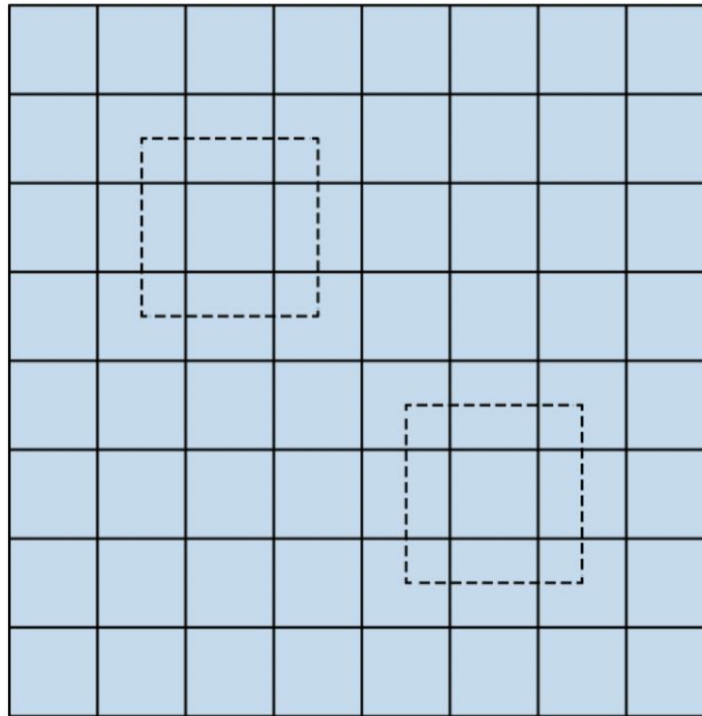
Multi-scale feature layers

- SSD overlays a grid of anchors (also called as priors) around the image
- SSDs fail to consider the objects near to the camera since it requires different scale of anchors
- Isolated horse feature

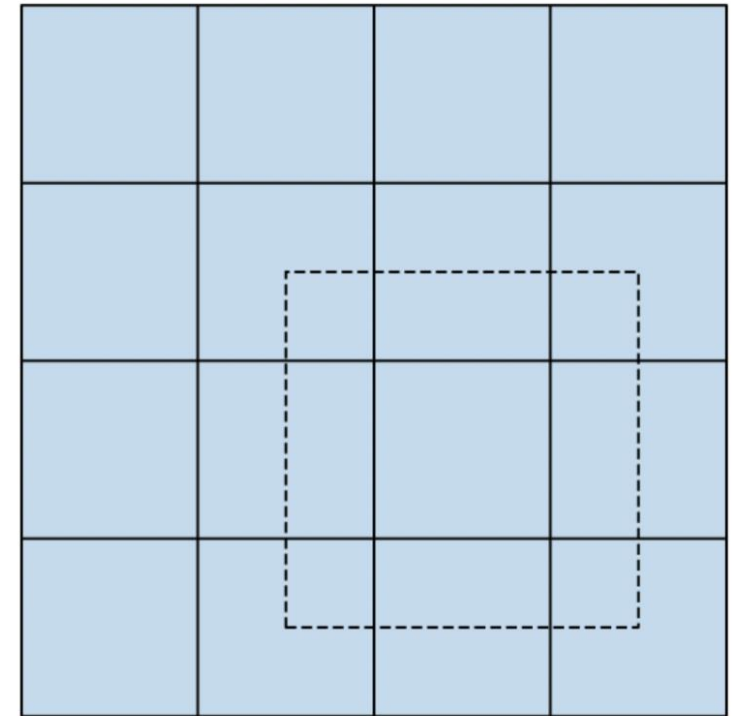


Multi-scale feature layers

**Lower-resolution feature maps
detect larger-scale objects (right);
higher-resolution feature maps
detect smaller-scale objects (left).**



8×8 feature map

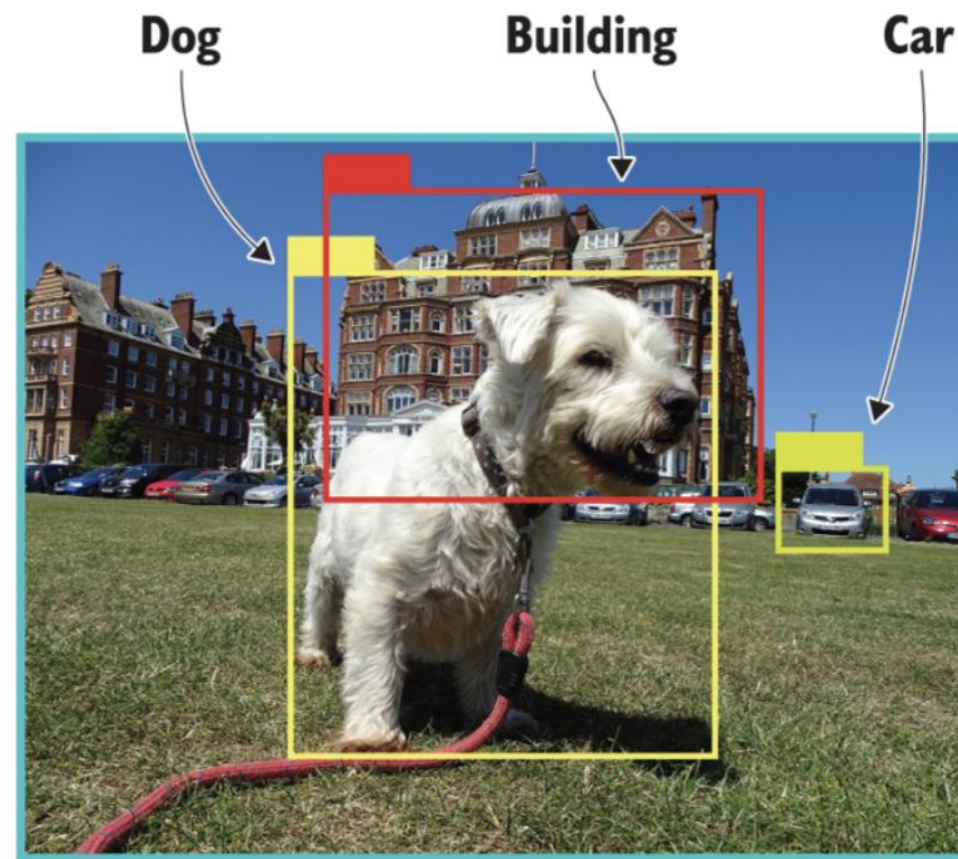
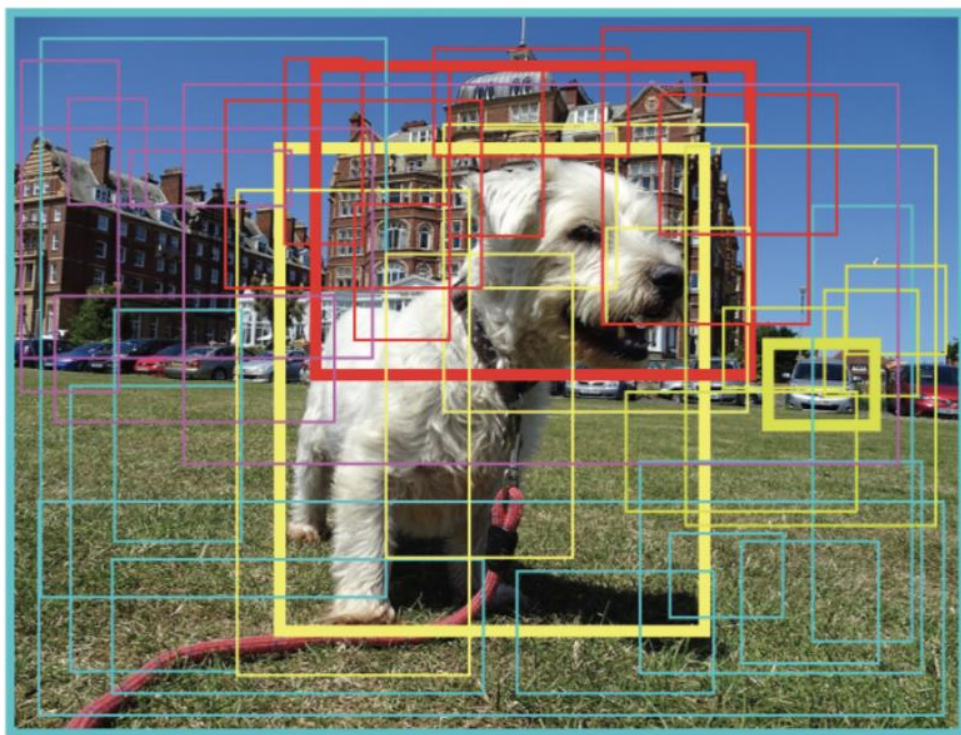


4×4 feature map

Effect of using multiple output layers

Prediction source layers from:						mAP use boundary boxes?		# boxes
conv4_3	conv7	conv8_2	conv9_2	conv10_2	conv11_2	Yes	No	
✓	✓	✓	✓	✓	✓	74.3	63.4	8,732
✓	✓	✓	✓	✓		74.6	63.1	8,764
✓	✓	✓	✓			73.8	68.4	8,942
✓	✓	✓				70.7	69.2	9,864
✓	✓					64.2	64.4	9,025
	✓					62.4	64.0	8,664

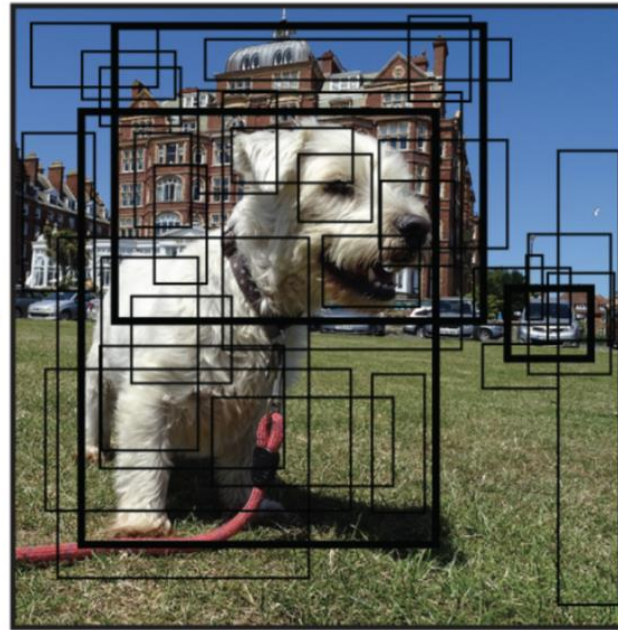
Non-maximum suppression



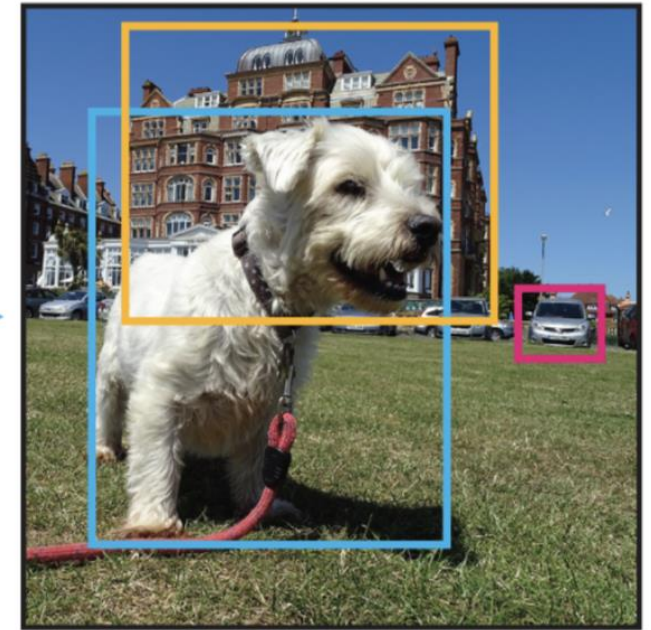
You Look Only Once (YOLO)



Splits the image into grids



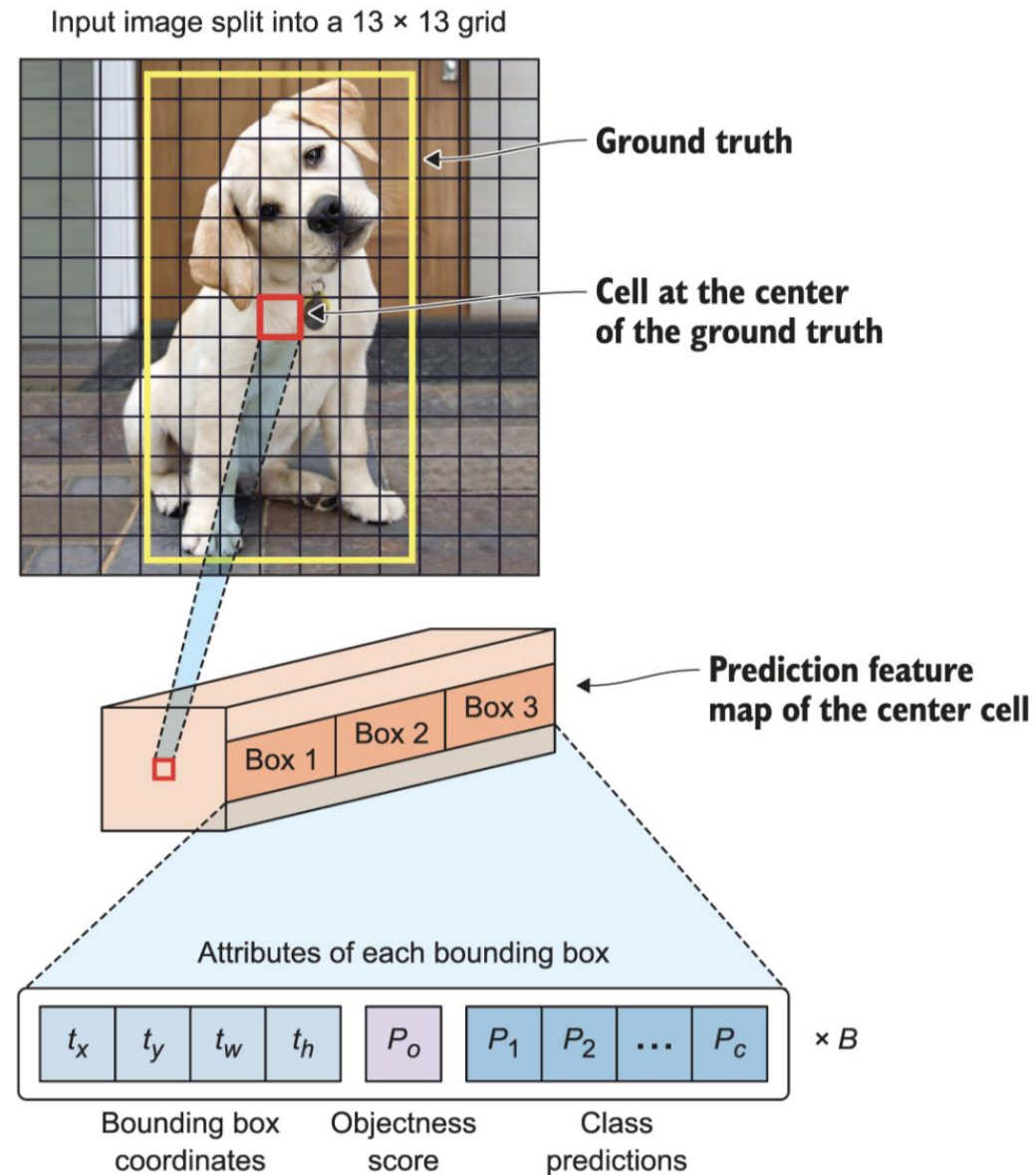
Predicts bounding boxes
and classifications



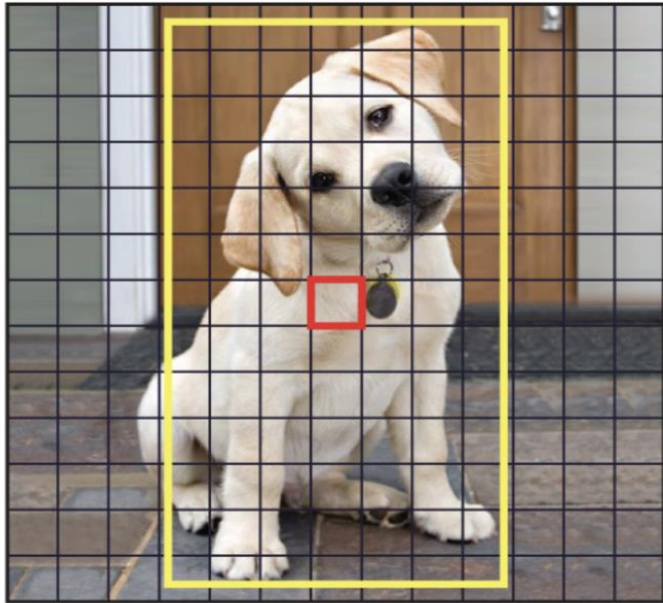
Final predictions after
non-maximum suppression

YOLO workflow

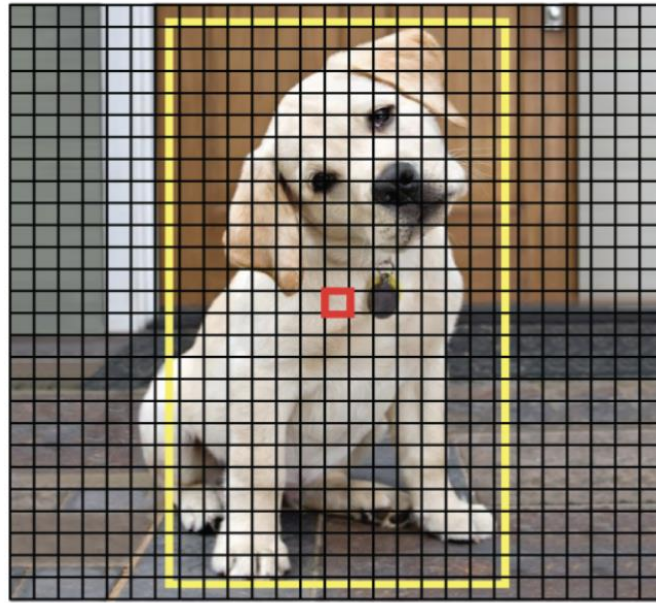
- Example of a YOLOv3 workflow when applying a 13×13 grid to the input image.
- The input image is split into 169 cells.
- Each cell predicts B number of bounding boxes and their objectness score along with their class predictions.
- In this example, we show the cell at the center of the ground-truth making predictions for 3 boxes ($B = 3$).
- Each prediction has the following attributes: bounding box coordinates, objectness score, and class predictions



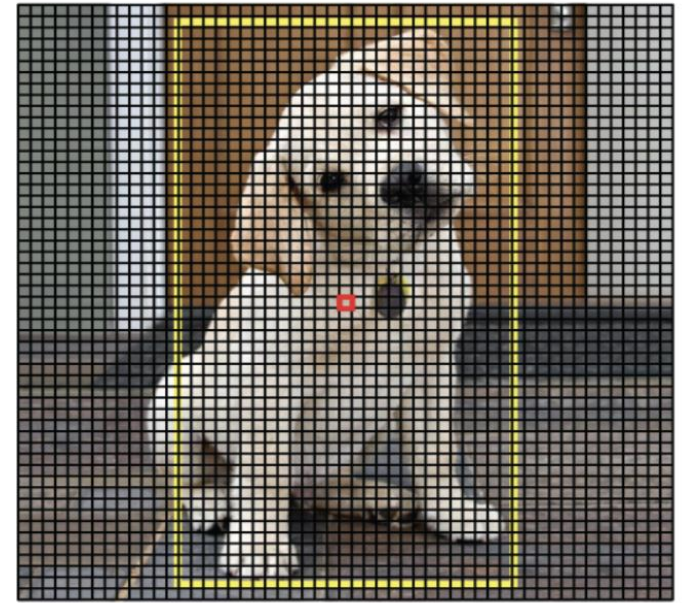
Prediction feature maps at varied scales



13×13

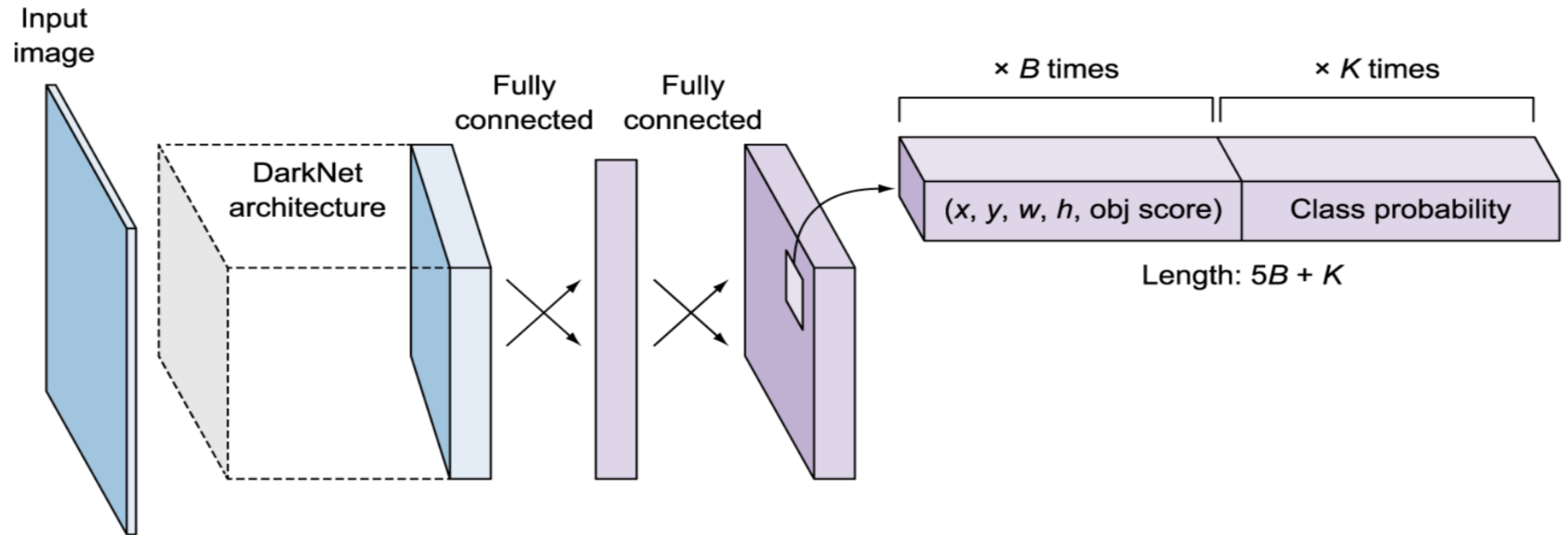


26×26

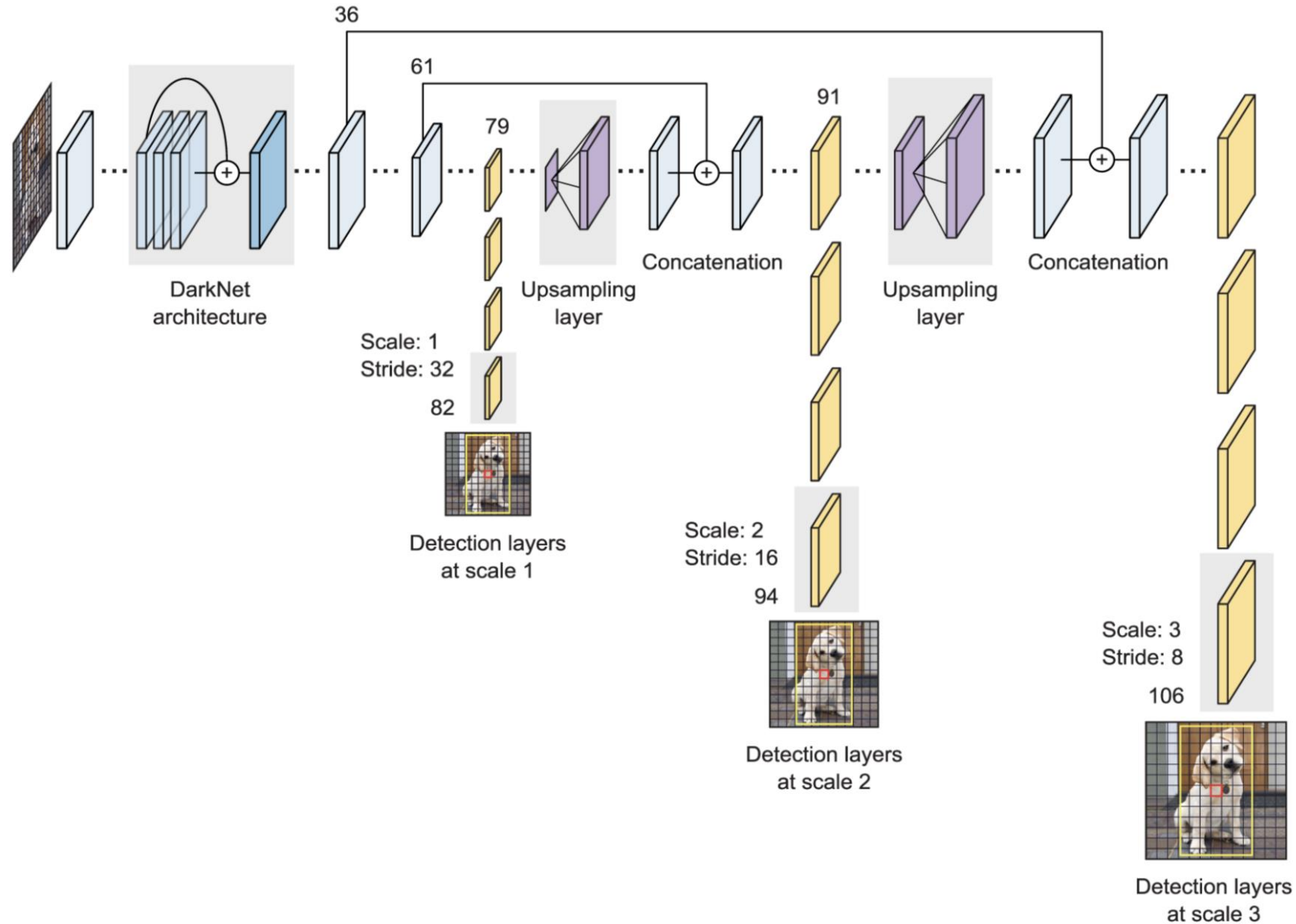


52×52

YOLO Architecture



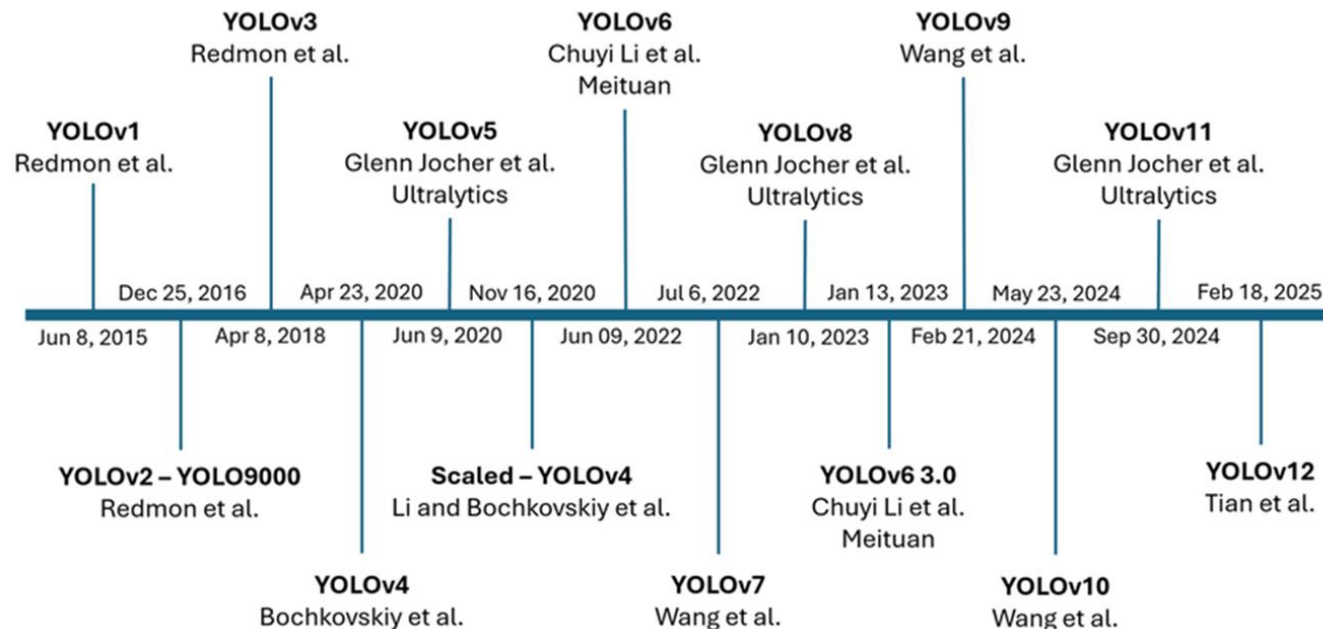
YOLO Network Architecture



YOLO model versions

You only look once -> Single stage object detection model

- YOLO predicts boxes + classes in one forward pass, enabling real-time inspection speeds.
- Good accuracy–efficiency tradeoff: YOLO offers lightweight variants suitable for factory/edge deployment while keeping strong accuracy.
- A family of versions (v1 → v12+): newer releases refine feature extraction, optimization, and efficiency



YOLO versions

Version	Year	Key Developer	Main Innovation
YOLOv1	2015	Joseph Redmon	Single-shot detection grid vison+1
YOLOv2	2016	Joseph Redmon	Anchor boxes, batch norm yolovx+1
YOLOv3	2018	Joseph Redmon	Multi-scale, Darknet-53 kili-technology+1
YOLOv4	2020	Alexey Bochkovskiy	CSPDarknet, better accuracy vison+1
YOLOv5	2020	Ultralytics	PyTorch, easy training yolovx
YOLOv8	2023	Ultralytics	Segmentation, pose support huggingface+1
YOLOv10	2024	Tsinghua Univ.	NMS-free, edge-optimized yolovx
YOLOv11	2024	Ultralytics	Hybrid CNN-transformer vison+1
YOLO26	2026	Ultralytics	Edge deployment, NMS-free inference blog.roboflow+1

Performance Comparison with Metrics

Aspect	Early (v1-v4)	Modern (v8+)	YOLO26
Speed	Moderate	High	Edge-optimized
Accuracy (mAP)	60-70%	80-90%	SOTA on COCO
Size	Large	Compact	Smaller models
Tasks	Detection	+Seg/Pose	All + OBB

Reference

- Book – Deep Learning for Vision Systems by Mohamad Elgendy, Manning Publication, November 2020